

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL2 - CONCEPT MAPPING

Course Code:	CSA1205	Course Title:	Computer Architecture for Multicore Systems
CO Assessed:	CO1	Assessment:	ASSESSMENT TOOL 2- CONCEPT MAPPING
Weightage:	35%	Total Marks:	25
CO1 Statement:	Analyze the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking (BL4)		
Student Name:		Reg.No.:	
Department:		Date:	

Instruction to Students

1. Answer two questions as per the Instruction.
2. Each question carries 12.5 mark.
3. Only the appropriate Tools can be used
4. Time allotted for the exam is 60 min

ANNEXURE A

2. Develop a comprehensive concept map linking the instruction execution cycle (fetch, decode, execute) with CPU performance parameters such as CPI, clock cycles, and execution time. Show how each stage contributes to overall execution time, and include connections to factors like memory access, instruction complexity, and pipeline stalls. Also, represent how benchmarking is used to evaluate performance.
3. Create a concept map comparing 8085 and 8086 architectures, including their instruction sets, register organization, addressing modes, and memory segmentation (in 8086). Clearly show how these architectural features influence program execution, multitasking capability, and efficiency in real-time applications.

Code of Conduct:

I Bharathi.Kadiveti(192412132) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MARKS ALLOCATION

Criteria	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)
Q1					
Q2					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence

Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand
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Question no. (2) :- Concept Map: Instruction Execution Cycle and CPU Performance

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19/2/2022

Slot - D

CPU Performance

CLOCK CYCLES

- Number of cycles taken to execute an instruction

CPI (cycles / instr)

- Average no of cycles per instruction

Execution time

- Total time taken to execute a program

INSTRUCTION EXECUTION CYCLES

1. FETCH

Reads instruction from memory

2. DECODE

Control unit interprets opcode

3. EXECUTE

ALU performs arithmetic / logical operation

Memory Access (Cache / RAM)

Register Read

Result stored in Register / memory

TOTAL CLOCK CYCLES

CPU Execution time formula

$$\text{Execution Time} = \text{Instruction Count} \times \text{CPI} \times \text{Clock cycle time}$$

Factors affecting Performance

MEMORY LATENCY

- Time taken to access memory

Higher CPI values

Instruction complexity

- Complex instruction requires more cycles

More cycles

CACHE MISSES

- Data not in cache causes access to main memory

Longer fetch time

PIPELINES STALLS

- Hazards, branches, resource conflicts

Delayed execution

Benchmarking programs

SPEED CPU BENCHMARKS

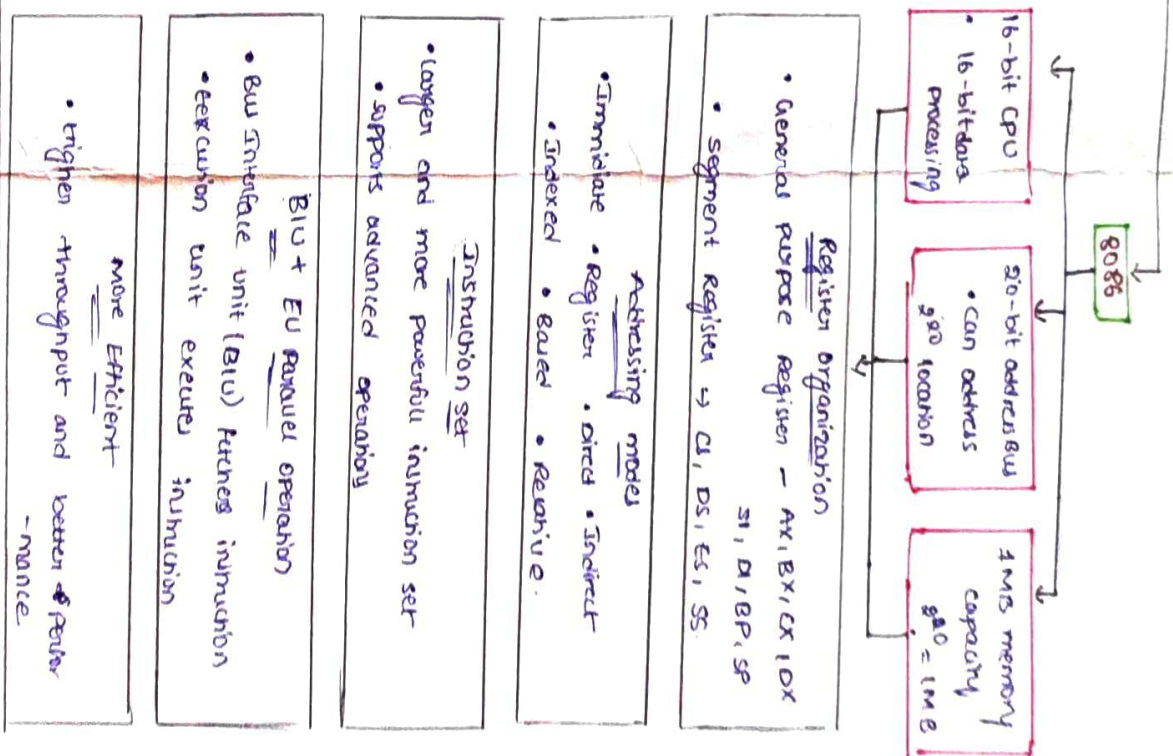
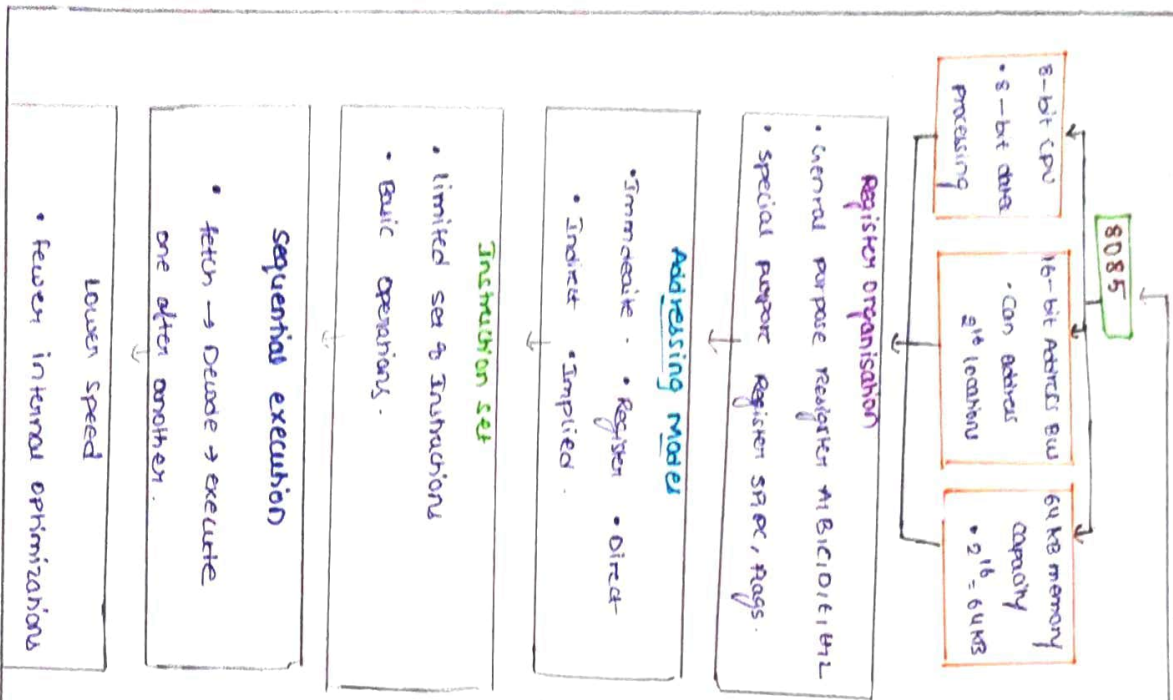
- Standard performance evaluation comparison benchmarks
- Measure real-world workloads

EMBEDDED BENCHMARK

- Used for embedded systems
- Focus on power efficiency, real-time

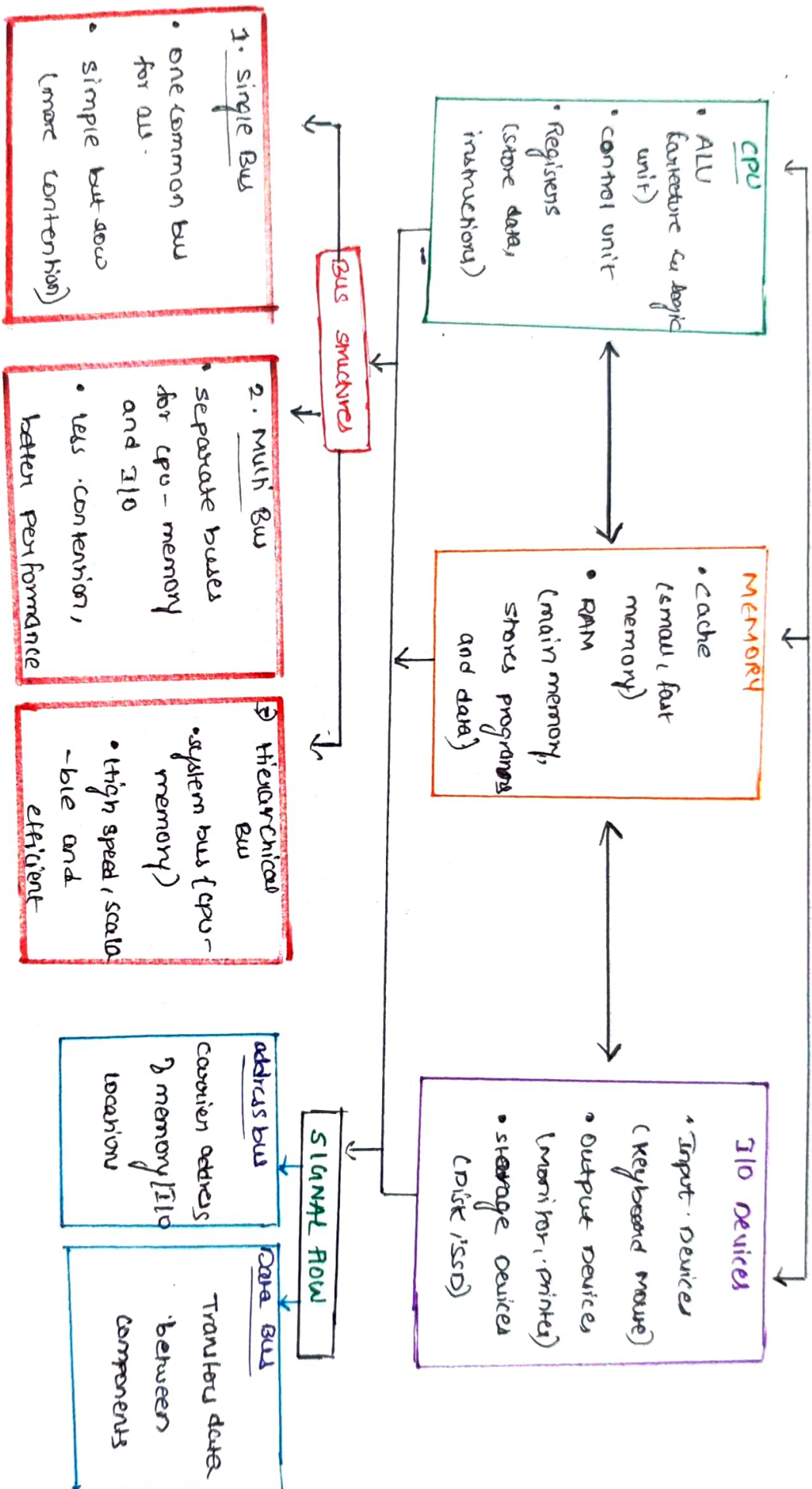
Question no - 3: Concept Map: Comparison of 8085 and 8086 Architecture

MICROPROCESSOR ARCHITECTURE



Ques 1:-

COMPUTER SYSTEM ARCHITECTURE



Result

efficient data and control flow, reduced delays, faster execution and better overall system performance.

CONCEPT MAPS

1) Number system

- Binary (base 2) $\rightarrow$ CPU understands only 0 and 1

- Hexadecimal (Base 16) $\rightarrow$

compact way to represent binary.

3) Instruction Representation

An instruction is a combination

opcode (operation) + operand

(data address) :

Example: ADD R1, [2000H]

←

←

Opcode

pend

(eg: Add) (eg, data|address)

(eg., data|address)

Instructions are converted to binary and stored in memory

- number system,
- Instruction Representation,
- memory storage &
- debugging.

4) Memory storage

CPU Interpretation:

- Instructions are stored in memory as binary (0 and 1)

←

- CPU fetches instruction from memory.

15.

- CPU decodes it (interprets operands and operations)

2) Data conversion :-

- Binary $\leftrightarrow$ Hexadecimal
- Group binary in 4 bits from right, convert each group to hex.

ex: binary $\rightarrow$ 1011 1100 0110

↩

19

11

3

2

Hex :- BCG

5) Debugging:

- During debugging, memory, content, register values, addresses and instructions are shown.

- Binary is hard to read.
- Hexadecimal is used because it is short, clear and easy to understand.

ADDRESSING MODES AND INSTRUCTION TYPE

